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Microbial necromass carbon enhances arsenic methylation in paddy soils

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Microbial necromass carbon (MNC) constitutes a critical component of soil organic carbon. Yet, how MNC regulates microbial arsenic (As) methylation processes in soil remains unclear. Across major Chinese rice-growing regions, bacterial and fungal necromass carbon showed significant positive correlations ($P < 0.05$) with the transcribed *arsM* gene. Soil incubation experiments with seven soils explored how straw and three types of MNC—gram-positive bacterial necromass carbon (G^+-NC), gram-negative bacterial necromass carbon (G^--NC), and fungal necromass carbon (F-NC)—affect As methylation. Our results demonstrated that all types of MNC enhanced As methylation, and G^--NC exhibiting the most pronounced effect on methylated As accumulation. The addition of 10 to 60 mg G^--NC maximally increased As(III) by 43.0 to 75.9% and enhanced methylated As by 4.4- to 18.0-fold in soil porewater vs. the control. Further, metagenomic and metatranscriptomic analyses demonstrated that G^--NC addition upregulated the relative abundance of transcribed *arsM* and *arsC2* genes, which were mostly assigned to *Acidobacteriota*, *Pseudomonadota*, *Planctomycetota,* and *Bacteroidota*. Notably, the transcriptional activity of *arsM*-harboring *Methanosarcinales* and *Moorellales* was markedly enhanced at the order level. By promoting As reduction process, G^--NC provides more substrates for As methylation process in soil. Furthermore, G^--NC could be used as a carbon source for As-methylating microorganisms, stimulating the transcriptional activity of *arsM*, which has been confirmed by the incubation experiment with pure culture of *Paraclostridium benzoelyticum* TC8. This study highlights the critical role of MNC in regulating As biogeochemistry, establishing a basis for predicting the extent of As methylation and risk of rice straighthead disease in paddy ecosystems.

arsenic | arsenic methylation | microbial necromass carbon | *arsM* gene | paddy soil

Significance

Microbial necromass carbon (MNC) is a key component of soil organic carbon, and methylated arsenic (As) critically affects rice yield. However, the role of MNC in regulating microbial As methylation remains unclear. We found that three MNC types enhance soil As reduction and methylation, with gram-negative bacterial necromass carbon (G^--NC) having the strongest effect on promoting As methylation. G^--NC promotes As reduction, while also serving as a carbon source for As-methylating microorganisms, stimulating *arsM* gene expression. Our findings establish MNC as a previously overlooked regulator in As biogeochemical cycling and reveal the pathway through which G^--NC influences As methylation. These observations provide a basis for predicting the accumulation of methylated As and the risk of straighthead disease in rice.

Arsenic (As) is a metalloid that poses considerable health risks in the environment due to its behemoth toxicity and wide abundance. In China, soil As concentrations increased from 11.9 mg kg^{-1} in 2000 to 12.6 mg kg^{-1} in 2020, resulting in 13.9% of the population living on soils that exceeded the 20 mg kg^{-1} threshold and facing significant health threats (1, 2). Therefore, the As migration and transformation in paddy soil have been extensively studied (3, 4). In paddy fields, inorganic As species, primarily arsenate [As(V)] and arsenite [As(III)], dominate. Under flooded conditions, As(V) is reduced to As(III), the substrate for methylation, through processes mediated by *arrA* genes (respiratory reduction) or *arsC* genes (intracellular reduction) (5, 6). Subsequently, As(III) is methylated stepwise by arsenite S-adenosylmethionine methyltransferase (ArsM), which is encoded by the functional gene *arsM*, using S-adenosylmethionine as the methyl donor. This process generates methylated species including monomethylarsenate (MMAs), dimethylarsenate (DMAs), and trimethylarsine (TMAs) in paddy soil (7, 8). The methylation process undoubtedly increases the accumulation of methylated As, which triggers rice straighthead disease and consequently impairs grain formation and reduces grain yield (9, 10). For instance, studies have shown that DMAs exposure significantly inhibits rice seed setting; the seed setting rate dropped dramatically to 19%, 11%, and 6% under 2, 5, and 10 μM DMAs treatments, respectively, underscoring its severe impact on reproductive development.

The As reduction and methylation process in paddy soils is strongly influenced by environmental factors such as redox potential (Eh), fertilizer application, and climate change (2, 11–13). Specifically, the input of organic matter, including straw, root exudates, and humic substances, profoundly influences the As transformation in paddy soils (14–16). For example, previous results showed that organic matter affected the release of As (17, 18), and As methylation efficiency increased by 57.4 to 260.1% in pig manure- or rice straw-treated paddy soils compared to the control (19). The input of such organic matter drives the As transformation due to the following reasons: 1) The decomposition of organic matter significantly lowers the soil Eh, establishing reducing conditions that favor anaerobic microbial

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The authors declare no competing interest.

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metabolisms essential for As(V) reduction and methylation (20). 2) Organic matter acts as electron donors and shuttles to promote As(V) reduction (16, 21). 3) Organic matter selectively enriches functional microorganisms associated with As reduction and methylation, such as sulfate-reducing bacteria (SRB), *Methanobacteria*, *Geobacter*, *Anaeromyxobacter*, and *Pseudomonas* (16, 20–23).

Microbial necromass carbon (MNC), mainly originating from dead microbial cell walls, constitutes a critical component of soil organic carbon (SOC) (24, 25). Hu et al. demonstrated that MNC contributes approximately 40% of global SOC (758 Pg C), surpassing plant-derived carbon inputs in the carbon pool (25). MNC is primarily composed of bacterial necromass carbon (B-NC) and fungal necromass carbon (F-NC). Generally, B-NC exhibits relatively high susceptibility to microbial decomposition, a characteristic potentially associated with its molecular architecture, specific chemical composition, and nitrogen-to-carbon ratio. Some studies have found that MNC can be absorbed on mineral surfaces and retained in the soil (26, 27). Yet, how MNC influences mineral dissolution, particularly the biotransformation of As, remains largely unexplored in paddy soil.

To investigate the role of MNC in regulating As methylation in paddy soil, the correlation between amino sugar content and the abundance of the *arsM* gene was quantified on a national scale. Based on this, soil incubation experiments were conducted to elucidate how MNC affects As methylation. Furthermore, using multiomics analysis (metagenomics/metatranscriptomics), functional microorganisms involved in As methylation upon MNC addition were explored.

Results

Correlations between MNC Contents and the Relative Abundance of *arsM* Gene in Paddy Soils. We collected 21 representative soil samples (three replicates each, n = 63) across the major rice-growing regions in China (Fig. 1*A* and *SI Appendix*, Table S1). Quantitative amino sugar analysis of 21 paddy soils revealed B-NC contents ranging from 356.6 to 2,219.6 mg kg^{-1} with a mean of 1261.3 mg kg^{-1}, while F-NC contents spanning 1,331.9 to 7,586.1 mg kg^{-1} with a mean of 4441.5 mg kg^{-1} (Fig. 1*B* and *SI Appendix*, Table S2). Significant positive correlations were observed between the relative abundance of transcribed *arsM* gene and both B-NC and F-NC (Fig. 1 *E* and *F*), whereas no significant correlations were found between MNC (B-NC and F-NC) and the relative abundance of transcribed *arsC* gene (Fig. 1 *C* and *D*). This field-scale association suggests microbial necromass may serve as a potential regulator of the As methylation process.

Effects of Straw and Different MNC Types on As Reduction and Methylation in Paddy Soils. We conducted soil incubation experiments with plant residue (rice straw) and three MNC types (F-NC, G^+-NC, and G^--NC) across 7 paddy soils (KY, WZ, GZ, MG, HY, TC, GL). Compared with control treatment (ranged from 0.58 to 128.21 μg L^{-1}), addition of G^+-NC (10.00 to 434.89 μg L^{-1}, $P < 0.05$), and G^--NC (12.01 to 459.78 μg L^{-1}, $P < 0.05$) both significantly increased As(III) concentration (Fig. 2*A* and *SI Appendix*, Table S3). Regarding DMAs in porewater, their concentrations varied significantly depending on the type of MNC. Compared to the control treatment (0.05 to 2.82 μg L^{-1}), significant increases were observed in the F-NC (0.19 to 5.40 μg L^{-1}, $P < 0.05$), G^+-NC (0.73 to 5.41 μg L^{-1}, $P < 0.01$), and G^--NC (0.85 to 7.18 μg L^{-1}, $P < 0.01$) treatments. Among these, the G^--NC treatment demonstrated the most pronounced effect, being significantly higher than all other treatments (Fig. 2*C* and *SI Appendix*, Table S5). In contrast, no significant difference in MMAs concentration was observed among the five treatments: control (0.00 to 0.84 μg L^{-1}), straw (0.00 to 3.92 μg L^{-1}), F-NC (0.00 to 6.03 μg L^{-1}), G^--NC (0.00 to 7.82 μg L^{-1}), and G^+-NC (0.00 to 2.31 μg L^{-1}) (Fig. 2*B* and *SI Appendix*, Table S4).

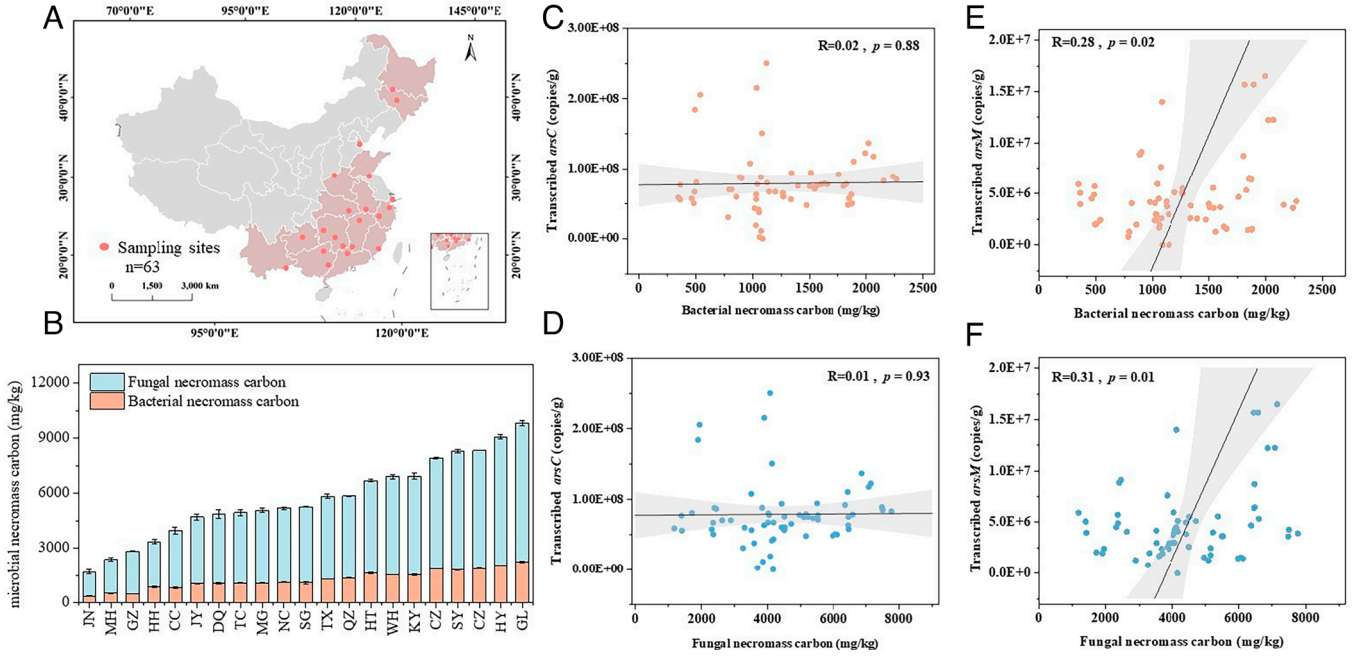


Fig. 1. Geographic locations of 21 paddy soils and the correlation between MNC and the relative abundances of transcribed *arsC* and *arsM* genes. (*A*) Geographic distribution of the 21 sampling sites. Each circle represents one site (n = 3 replicates per site). (*B*) Contents of bacterial and fungal necromass carbon in the 21 paddy soil samples. Data are presented as mean ± SD (n = 3). Linear regression analysis of the *arsC* gene abundances (RNA level) with bacterial necromass carbon (*C*) or fungal necromass carbon (*D*). Linear regression analysis of the *arsM* gene abundances (RNA level) with bacterial necromass carbon (*E*) or fungal necromass carbon (*F*).

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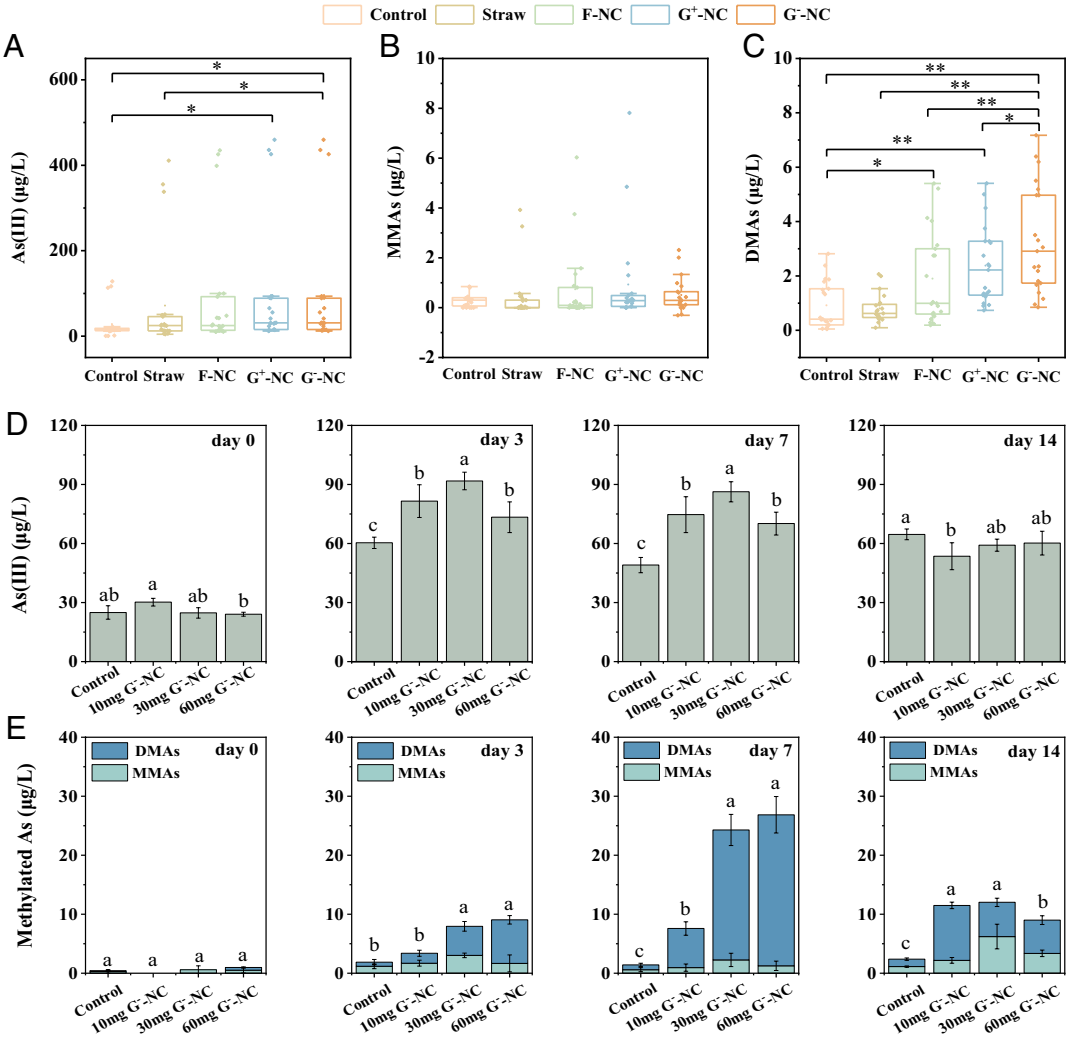


Fig. 2. The concentrations of As(III), MMAs, and DMAs are altered by MNC amendments, with the most pronounced increase in methylated As driven by the G⁻-NC. Concentrations of As(III) (*A*), MMAs (*B*), and DMAs (*C*) in porewater across seven soils on day 7 of incubation with straw and different MNC additions. Dynamics of As(III) (*D*) and methylated As (MMAs and DMAs) (*E*) in Experiment B with varying G⁻-NC doses. In (*A*–*C*), asterisks indicate significant differences from the control at each time point as determined by *t* test (*$P < 0.05$, **$P < 0.01$). Significant differences between treatments in (*D* and *E*) were determined by ANOVA. Notes: Control (no amendment), and G⁻-NC additions of 10 mg (0.5 g C kg^{-1}), 30 mg (1.5 g C kg^{-1}), or 60 mg (3.0 g C kg^{-1}) per vial. Data are presented as mean ± SD (n = 5). The box plots in (*A*–*C*) represent the 25th to 75th percentiles (median line), with whiskers showing the min and max values.

Dynamics of As(III) and Methylated As in Paddy Soils with Amended With Different Doses of G⁻-NC. The addition of 10 to 60 mg of G⁻-NC significantly promoted the production of As(III) and methylated As. The addition of G⁻-NC (10 to 60 mg) increased the concentration of As(III) in porewater by 21.6 to 52.1% and 43.0 to 75.9% at day 3 and day 7, respectively (Fig. 2*D* and *SI Appendix*, Table S6). The addition of 30 and 60 mg G⁻-NC significantly elevated methylated As concentrations by 3.2-fold (30 mg) and 3.8-fold (60 mg) on day 3. The addition of G⁻-NC (10 to 60 mg) increased methylated As by 4.4 to 18.0-fold at day 7 and by 2.8 to 4.0-fold at day 14 (Fig. 2*E* and *SI Appendix*, Table S7).

The Relative Abundance of the *arsM*, *arsC*, *arrA*, and *mcrA* Genes in the Metagenome and Metatranscriptome. To assess the effect of G⁻-NC amendment, we performed metagenomic and metatranscriptomic analyses on 10 samples (five amended with 30 mg of G⁻-NC and five control samples) collected on day 7 from Experiment B. Metagenomic results revealed no significant difference in *arsM* gene abundance between G⁻-NC-treatment and control (1.13 ± 0.08 vs. 1.16 ± 0.05 RPKG, Fig. 3*A*). However, the relative abundance of transcribed *arsM* genes exhibited a marked increase in G⁻-NC-amended soils (29.34 ± 6.40 vs. 22.57 ± 1.87, $P < 0.05$). Similarly, the relative abundances of the *arsC2* genes in the metagenome showed no significant difference between G⁻-NC-treated and control groups. In contrast, a significant increase in relative abundance of transcribed *arsC2* genes was observed in G⁻-NC-treatment samples compared to controls (17.15 ± 1.37 vs. 11.30 ± 0.59, $P < 0.05$). In contrast, both the relative abundance of *arsC1* gene (0.22 ± 0.02 vs. 0.22 ± 0.01) and its transcripts (0.56 ± 0.29 vs. 0.96 ± 0.48) showed no significant difference between G⁻-NC-treatment and control. Notably, the relative abundances of *mcrA* genes were significantly elevated in both the metagenome (0.14 ± 0.01 vs. 0.04 ± 0.01 RPKG) and the metatranscriptome (994.35 ± 236.05 vs. 7.52 ± 3.46) upon G⁻-NC addition ($P < 0.05$). Although there was no significant difference in the relative abundance of *arrA* genes in the metagenome between the control and G⁻-NC-amended soils, the relative abundance of transcribed *arrA* genes was significantly higher in the control soils than in the amended soils (0.05 ± 0.03 vs. 0.00 ± 0.01, $P < 0.05$).

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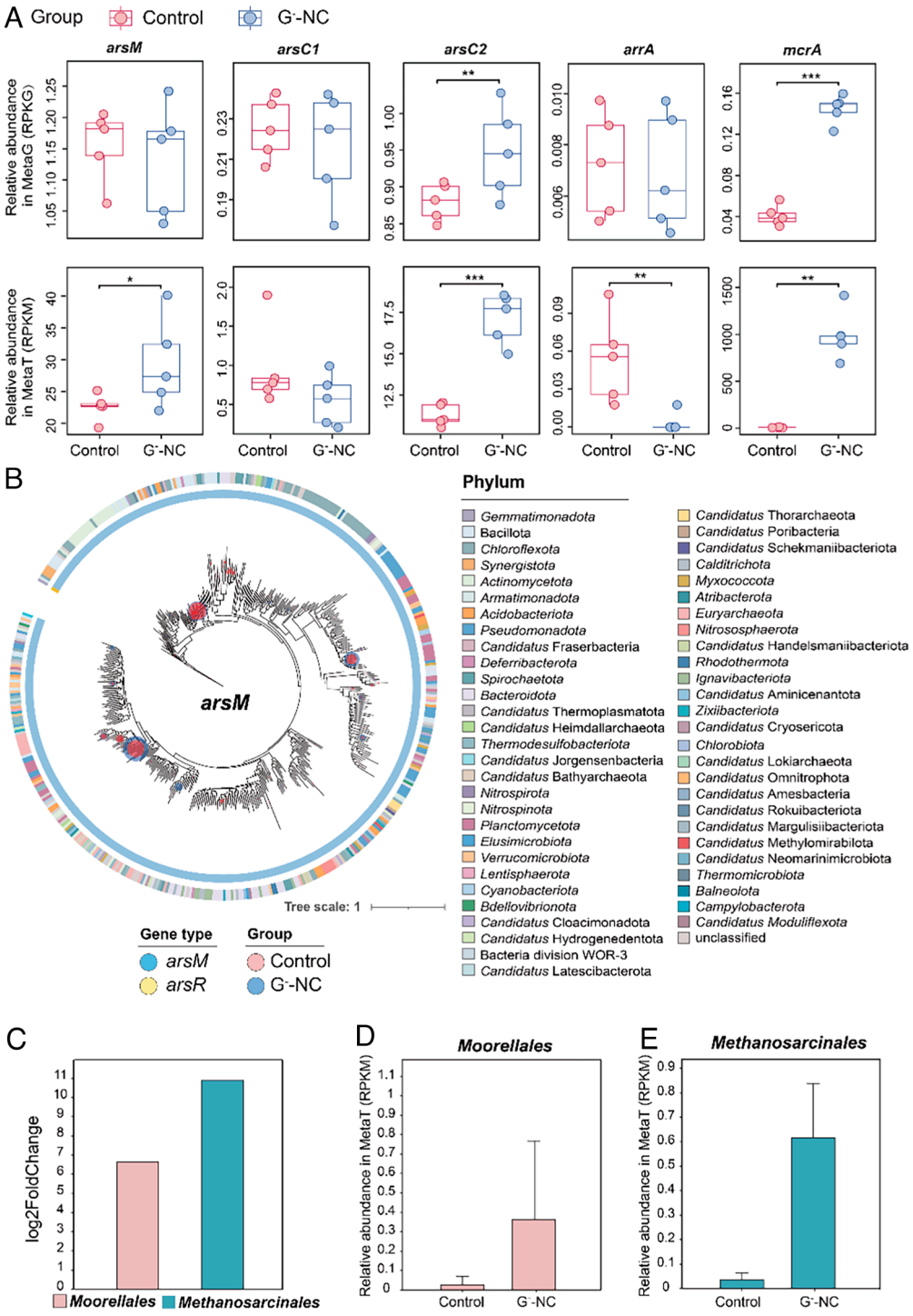


Fig. 3. Abundance and expression of *arsM*, *arsC*, *arrA,* and *mcrA* genes, and taxonomic classification of *arsM* subcommunities in control and G⁻-NC paddy soils. (*A*) Relative abundance of *arsM*, *arsC*, *arrA,* and *mcrA* genes derived from metagenomic and metatranscriptomic datasets in control (pink) and G⁻-NC paddy soils (blue). (*B*) Phylogenetic placement of *arsM* gene-carrying reads identified in metatranscriptomes from control (red bubble) and G⁻-NC (blue bubble) paddy soils. In the phylogenetic tree, the outer ring's distinct colors denote different phyla, while the inner ring's different colors signify different genes: the *arsM* gene (in blue) and the outgroup *arsR* gene (in yellow). (*C*–*E*) Log2fold change of *arsM* gene-carrying reads identified in metatranscriptomes that were significantly transcribed between control and G⁻-NC paddy soils (negative binomial test, adjusted $P < 0.05$).

In the metatranscriptomic data, 96.0% of the *arsM* gene reads were successfully mapped to the phylogenetic tree, corresponding to 55 phyla. The transcribed *arsM* subcommunities were predominantly composed of the *Acidobacteriota*, *Pseudomonadota*, *Planctomycetota*, *Bacteroidota*, *Cyanobacteriota*, *Chloroflexota*, and *Bacillota,* which represented 65.5% of the total *arsM* genes (Fig. 3*B* and *SI Appendix*, Table S8). DESeq2 analysis revealed statistically significant changes in the relative abundance of genes identified in the metatranscriptomic data with or without addition of G⁻-NC. At the order level, *Moorellales* and *Methanosarcinales* were significantly upregulated under G⁻-NC treatment (Fig. 3*C*). In addition, the relative abundance of *Moorellales* and *Methanosarcinales* in metatranscriptomics was significantly increased by 13.1 and 16.6 times after the addition of G⁻-NC, respectively (Fig. 3 *D* and *E*).

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Identification of Metagenome-Assembled Genomes (MAGs). A total of 43 MAGs with ≥70% completeness and ≤10% contamination were reconstructed from 10 metagenomic datasets (*SI Appendix*, Table S9). Among these, 41 MAGs were affiliated with bacterial phyla including *Bacillota*, *Patescibacteria*, *Chloroflexota*, *Verrucomicrobiota*, *Gemmatimonadota*, *Bacteroidota*, *Myxococcota*, *Pseudomonadota*, and *Acidobacteriota*, while 2 archaeal MAGs belonged to *Thermoproteota* and *Nanoarchaeota* (Fig. 4). Notably, 76.7% (33/43) of MAGs harbored *arsM* genes, including 1 archaeal MAG (*Thermoproteota*) and 32 bacterial MAGs distributed across *Bacillota*, *Chloroflexota*, *Verrucomicrobiota*, *Gemmatimonadota*, *Bacteroidota*, *Pseudomonadota*, and *Acidobacteriota*. Analysis of the metatranscriptomic data revealed that the *arsM* gene was transcribed in 97.0% (33/34) of the *arsM*-carrying MAGs. Notably, in 6 of these MAGs, the relative abundance of transcribed *arsM* was significantly higher ($P < 0.05$) in G^--NC treatment soils compared to the control soils. These *arsM*-carrying MAGs were distributed in the phyla of *Actinobacteriota*, *Bacteroidota*, *Chloroflexota,* and *Thermoproteota*.

Discussion

Our field survey of paddy soils revealed a significant positive correlation between MNC and the transcribed *arsM* gene ($P < 0.01$) (Fig. 1 *E* and *F*), suggesting that MNC is a key factor regulating the As methylation process in paddy soil. To validate this

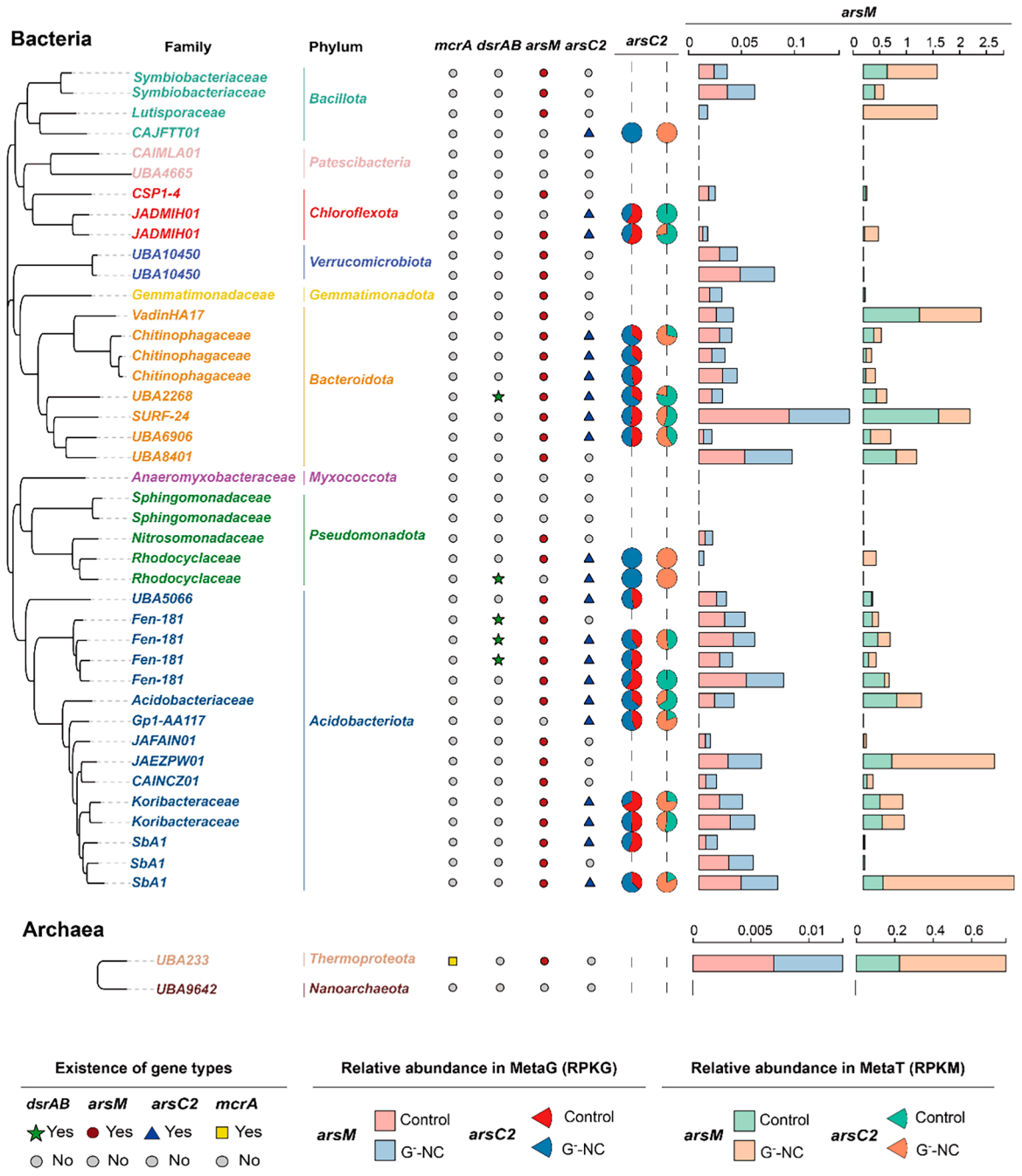


Fig. 4. Presence and expression of *arsM* and *arsC2* genes in MAGs. Clades of the phylogenetic tree are colored according to phylum and named by family, with the corresponding phylum classifications shown on the right. The pie charts and bar plots represent the relative abundance of annotated As resistance genes (bar plots represent *arsM* genes and pie charts represents *arsC2* genes) based on metagenomic and metatranscriptomic data. The presence of the *mcrA*, *dsrA* or *dsrB*, *arsM*, and *arsC2* genes is indicated by colored symbols.

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hypothesis, microcosm experiments were conducted comparing MNC (F-NC, G^+-NC, and G^--NC) with plant-derived carbon (rice straw) and controls. The results showed that MNC addition significantly increased concentration of DMAs compared to both straw-amended and control treatments, with G^--NC exhibiting the strongest promotion effect on As methylation (Fig. 2 *C* and *E* and *SI Appendix*, Fig. S1). Moreover, during culture of the strain *Paraclostridium benzoelyticum* TC8 capable of As methylation, the addition of G^--NC as the sole carbon source accelerated DMAs production while significantly enhancing CO_2 release (*SI Appendix*, Fig. S2). Consequently, G^--NC served as a direct carbon source for TC8, which was isolated from TC soil. Labile carbon sources activate the biosynthetic pathway of methyl donors like S-adenosylmethionine (SAM), thereby enhancing the functional expression of arsenite methyltransferase (ArsM) (19). G^--NC exhibits a much higher release of labile organic carbon compared to G^+-NC, primarily due to inherent differences in overall cellular architecture. Gram-positive bacteria possess a thick, multilayered peptidoglycan cell wall that is difficult to degrade and restricts the leakage of cellular contents. In contrast, the thin, single-layered peptidoglycan cell wall of Gram-negative bacteria is more prone to rupture, facilitating the release of cytoplasmic compounds (28). Moreover, our results showed that the dissolved organic carbon (DOC) and dissolved organic nitrogen (DON) released from G^--NC was significantly higher than that from both G^+-NC, FNC, and straw (*SI Appendix*, Fig. S5). Notably, in this study, plant-derived carbon (straw) was included as a realistic and agronomically relevant control to benchmark the influence of MNC. Like other plant residues, straw must be extensively decomposed first, yet the significantly stronger promoting effect observed with G^--NC compared to straw underscores the importance of microbial-derived necromass in regulating As transformation, highlighting that not all organic carbon inputs are equivalent in their capacity to drive methylation processes.

The methylation of As, a key process determining its mobility and toxicity in paddy ecosystems, has direct implications for the incidence of straighthead disease and rice grain yield. Understanding the drivers of this process is therefore critical for developing strategies to manage As bioavailability. It is widely accepted that As methylation occurs via a pathway in which As(III) is methylated by the S-adenosylmethionine methyltransferase (ArsM), which is encoded by the functional gene *arsM*. Our result showed that G^--NC facilitates As(V) reduction to As(III) by decreasing Eh (*SI Appendix*, Fig. S3) and upregulating arsenate reductase genes (*arsC2*) (Fig. 3*A*), ensuring sufficient substrate availability for subsequent methylation (16). Furthermore, based on metatranscriptomic analyses of 10 samples, the addition of G^--NC significantly enhanced the relative abundance of the transcribed *arsM* gene compared to that in the control group (Fig. 3*A*). This enhancement was accompanied by a marked increase in the relative transcriptional abundance of key *arsM*-harboring groups, specifically *Moorellales* and *Methanosarcinales* (Fig. 3*C*). Meanwhile, under G^--NC treatment, the relative abundance of the transcribed *mcrA* gene, which is crucial for methanogenesis, was significantly increased (Fig. 3*A*). Furthermore, a strong correlation ($P < 0.05$) was observed between transcribed *arsM* and *mcrA* genes (*SI Appendix*, Fig. S6). We therefore propose that these two orders (*Moorellales* and *Methanosarcinales*) played a major role in mediating As methylation in the paddy soil amended with G^--NC. In addition, previous studies have reported that the enrichment of *Methanosarcinales* can release intermediate metabolites during methane production to provide additional electron donors for As-methylating bacteria (29).

Traditionally, SRB have been considered the dominant microbial group responsible for As methylation in flooded paddy soils (30). However, our MAGs data revealed that under G^--NC amendment, the relative abundance of transcribed *arsM* gene in MAGs annotated as SRB did not differ significantly from the control (0.67 vs. 0.87). In contrast, we observed a significantly greater relative abundance of the transcribed *arsM* gene among methanogen-associated MAGs compared to the control (0.55 vs. 0.22, $P < 0.05$) (*SI Appendix*, Table S9). These findings suggest that although SRB contribute partially to As methylation, G^--NC amendment primarily enhances As methylation by stimulating the activity of methanogens (such as *Methanosarcinales* or *Thermoproteota*) (Fig. 3 *C* and *E*). Our BES inhibition experiment further supports this mechanism, as under coamendment with G^--NC and BES, methanogenic activity was suppressed, leading to a markedly reduced level of methylated As—significantly lower than that in the G^--NC-alone treatment and comparable to the control (*SI Appendix*, Fig. S4). Therefore, we propose that G^--NC promotes As methylation predominantly through enhancing methanogenic activity. Liang et al. also demonstrated that *Methanosarcina acetivorans* can catalyze the methylation of As(III) via the cytoplasmic arsenite methyltransferase (ArsM) during the stationary growth phase (31). Both the ArsC-catalyzed reduction of As(V) and the ArsM-catalyzed methylation of As(III) preferentially utilize thioredoxin (Trx) encoded by MA4683 as the physiological electron donor, establishing a redox link between methanogenesis and As transformation.

With the enhanced As reduction by G^--NC, more substrates are provided for the As methylation process. Furthermore, G^--NC promotes As methylation in paddy soil by supplying carbon sources to As-methylating microorganisms and enhancing the activity of key *arsM*-harboring taxa such as *Moorellales* and *Methanosarcinales*. In conclusion, our study identifies MNC as a key driver of As methylation. This insight shifts the perspective from focusing solely on external organic carbon inputs to emphasizing the management of the internal SOC pool. This finding provides a perspective for controlling As methylation and mitigating rice straighthead disease in paddy soils.

Materials and Methods

Soil Samples Collection and Amino Sugar analysis. A total of 63 soil samples were collected from 21 rice-growing regions across China after the rice harvest in 2022 to 2023, with three replicates per site (Fig. 1*A*). The topsoil (0 to 20 cm) was placed into sterile self-sealing bags immediately after harvest and transported to the laboratory on ice for basic soil physicochemical analysis, RNA extraction, and amino sugar determination. The methods of soil collection, pretreatment, and preservation referred to previous studies (15). The other part of the soils was air-dried and passed through a 2-mm sieve for a subsequent incubation experiment. The quantification of soil amino sugars was performed following the gas chromatographic methodology described by prior research (32). Soil samples were subjected to acid hydrolysis with 6 M HCl (8 h at 105 °C), followed by filtration, drying, and dissolution in 20 mL deionized water with subsequent neutralization to pH 6.6 to 6.8, before being centrifuged (3,000 rpm for 10 min) and dried by rotary evaporation. The residues underwent methanol dissolution, solvent evaporation under nitrogen gas at 45 °C, and then freeze-drying. Amino sugars were determined using an Agilent 6890A gas chromatograph equipped with flame ionization detection and HP-5 capillary column. Four amino sugar fractions were quantified: glucosamine (GlcN), galactosamine (GalN), muramic acid (MurA), and mannosamine (ManN).

Preparation of Microbial Necromass and Straw. Microbial necromass was derived from common soil microorganisms: *Fusarium oxysporum* (fungi), *Rhodococcus biphenylivorans* TG9 (Gram-positive bacteria), and *Escherichia coli* A2 (Gram-negative bacteria). Strains TG9 and A2 were cultured in LB medium. A 1

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mL aliquot of a preculture was inoculated into 1,000 mL of sterile LB medium. The cultures were subsequently incubated in a 30 °C shaking incubator for 24 h until reaching an optical density (OD600) of approximately 1.0. Bacterial cells were harvested by centrifugation (6,000 rpm, 10 min) followed by three-time washing with phosphate-buffered saline to remove residual medium components. The resulting cells underwent triple sterilization cycles via autoclaving (121 °C, 20 min per cycle) to obtain the bacterial necromass. The sterilized necromass was lyophilized, homogenized through mechanical grinding, and stored as fine powder for subsequent experiments. For fungi, PDA medium was used. Fungi were inoculated on the surface of the medium and then inverted in a 30 °C incubator for 5 days, and harvested with a spatula (33). The remaining steps were similar to those for bacterial necromass. The rice straw used as an amendment was collected from paddy fields in Hangzhou City, Zhejiang Province, China, and the variety was "Liangyou 3108." After harvesting the rice, it was placed in a 105 °C oven for 30 min (as a pretreatment step) and then kept in a 60 °C oven until completely dry. Subsequently, the rice straw was ground through a 100-mesh sieve for use in the subsequent experiments.

Soil Incubation Experiment.

Experiment A. Effects of different MNC types on As reduction and methylation in seven paddy soils. This experiment was designed to systematically investigate the effects of different types of MNC on As reduction and methylation in paddy soils. Seven soils (KY, WZ, GZ, MG, HY, TC, and GL) were used and the details of the soil sampling points were provided in the supplementary material *SI Appendix*, Table S1. Anaerobic incubation was conducted under laboratory conditions. Five treatments were designed, including Control (without MNC, CK), Straw (rice straw), F-NC (Fungal necromass carbon), G^+-NC (Gram–positive bacterial necromass carbon), and G^--NC (Gram–negative bacterial necromass carbon). After 20 g of soil was placed into 120 mL serum vials, 40 mL of deionized water was added to each. Prior to the formal experiment, the soils were flooded and preincubated for 7 d to reactivate the soil microorganisms. Each treatment had three replicates and was incubated at 25 °C in the dark. After 7 d of preincubation, Straw, FNC, G^+-NC and G^--NC were added or not added respectively, and the addition amount was 1.5 g C kg^{-1} soil (34). The serum vials were sealed and the headspace was purged with N_2 gas. Soil porewater was collected on day 7 of incubation for As speciation analysis. Portions of the porewater samples were acidified with concentrated HCL (100:1, v:v) to prevent transformation and filtered through 0.22 μm membrane filters before analysis. As species were quantified using HPLC-ICP-MS (35).

Experiment B. The dynamic process of As methylation affected by different contents of G^--NC in paddy soil was evaluated using TC soil. Four treatments including control (without G^--NC addition), 10 mg G^--NC (0.5 g C kg^{-1} soil), 30 mg G^--NC (1.5 g C kg^{-1} soil), and 60 mg G^--NC (3.0 g C kg^{-1} soil) were conducted in the laboratory. The experimental conditions were consistent with those in experiment A. Soil pore water was collected at days 0, 3, 7, and 14 after incubation for As speciation analysis. A pH/ORP meter (Leici 501, China) was used to measure the pH and redox potential (Eh) values at the soil–water interface. The concentration of DOC was determined using a Multi C/N TOC analyzer. The ferrous iron (Fe (II)) in soil porewater was analyzed using the 1,10-phenanthroline colorimetric method with a UV-VIS spectrophotometer at 510 nm (36).

Experiments C and D. To validate the findings from Experiments A and B, we examined in Experiment C whether *P. benzoelyticum* TC8–an As-methylating bacterium isolated from TC soil–could utilize G^--NC as its sole carbon source. In Experiment D, we evaluated the influence of methanogens on As methylation in paddy soil under G^--NC amendment, using the specific inhibitor sodium 2-bromoethanesulfonate (BES). Detailed methodologies are available in *SI Appendix*.

Quality Control, Assembly, Binning and ORF Prediction in MAGs. Metagenomic sequencing was performed on 10 samples (comprising control and 30 mg G^--NC treatments from Experiment B at day 7), followed by the recovery and analysis of MAGs. Quality control of the raw paired-end reads in both metagenomic and metatranscriptomic datasets was carried out by using multitrim v1.2.6 with the default settings (https://github.com/Kgerhardt/multitrim). The metatranscriptomic datasets had their rRNA reads removed using SortMeRNA v4.3.4 (37) (--paired_in). The assembled and coassembled contigs of the clean reads for each site's metagenomic data were generated using MEGAHIT v1.2.9 (38) (-min-contig-len 500). MAGs recovered from coassembled contigs were binned by BASALT v1.0.1 (39) retaining bins with >70 % completeness and <10 % contamination for analysis. The eligible MAGs dereplicated by dRep v3.3.0 (-sa 0.95-nc 0.30). The open reading frames (ORFs) from MAGs were predicted by Prodigal v2.6.3 (-p meta). Taxonomic annotation of these MAGs was performed using the "classify_wf" model from GTDB-Tk version 2.1.0. A phylogenetic tree based on MAGs was built with the "infer" tool in GTDB-Tk and visualized through TVBOT v2.6.0.

Functional Annotation, Identification, and Expression of *arsM*, *arrA*, *arsC* (*arsC1* and *arsC2*), and *mcrA* Genes. Short-read sequences of the *arsM*, *arrA*, *arsC1*, and *arsC2* and *mcrA* genes were annotated by DIAMOND BLASTx (40) (with the option "--sensitive") against the ROCker (4142, 43) database and McycDB. Subsequently, the best matches were sorted using the script BlastTab.best_hit_sorted.pl (publicly available at https://enve-omics.ce.gatech.edu/enveomics/docs?t = BlastTab.best_hit_sorted.pl). Afterward, the best matches were filtered either by ROCker model or by a set of parameters: identity greater than 80%, e-value less than 1×10^{-5}, and an alignment length of at least 25 amino acids. The predicted ORFs from the MAGs were searched against a As reference protein sequences database (https://github.com/SZ-MEH-Lab) via DIAMOND BLASTp. Only the best matches which have a minimum cut-off of a match of 60% identity, 70% query coverage and e-value of $1e^{-5}$ were considered as ArsM, ArrA or ArsC1 and ArsC2. The relative abundance of genes in the metagenomic datasets was calculated using RPKG [(the number of reads mapped to the gene)/(gene length in kb)/(genome equivalents)], which indicates the number of cells from those samples encoding the gene of interest (copies per cell). And RPKM [i.e., reads per kb per millions of reads, RPKM = (the number of reads mapping to gene)/(gene length in kb)/(the total number of reads after removal of rRNA reads)] means the relative abundance of transcribed genes in the metatranscriptomics. Briefly, the short-read metagenomes from each sample were mapped to MAGs via BLAT v2.3.4.1 (query length ≥100 bp and identity ≥ 95%), and the coverage of contig was calculated via BlastTab.seqdepth.pl (https://enveomics.ce.gatech.edu/enveomics/docs?t = BlastTab.seqdepth.pl).

Phylogenetic Placement of *arsM* Genes Carrying Reads. The ArsM tree (41) was used as a reference for the phylogenetic placement of metatranscriptomic reads identified as *arsM* genes through the ROCker Model. These placements were conducted using RAxML with the parameters "-f v -G 0.2 -m PROTGAMMALGX". The placement of the target protein-containing reads was visualized in iTOL v6 after processing the resulting visualization Newick format file generated by RAxML with JPlace.to_iTOL.rb (publicly available at https://envematics.ce.gatech.edu/enveomics/docs?t = Jplace.to_iToL.rb).

Statistical Analysis. All statistical analyses were performed using SPSS 22.0. Basic data processing and table creation were conducted in Microsoft Excel 2016. ANOVA was performed to analyze the effects of Straw, F-NC, G^+-NC, and G^--NC (with concentration gradients for G^--NC) on As(III) and methylated As (MMAs and DMAs). The linear regression analyses presented in Fig. 1 *C–F* were performed using the Linear Fit function within the "Simple Fit" tool of Origin 2021. This method is based on the ordinary least squares (OLS) algorithm to estimate the parameters of the linear model ($y = a + bx$). The p-value associated with the regression model was used to determine the statistical significance of the linear relationship. A *P*-value of less than 0.05 was considered statistically significant. The significance of differences in the relative abundance of functional genes between control and G^--NC was calculated via *t* test ($P < 0.05$) and visualized via the ggsignif software package. Differentially abundant taxa harboring the *arsM* gene across control and treatment groups were identified using DESeq2 v1.44.0 (44). All analyses and plots were performed in Origin 2022, R v4.2.3 (45), tvBOT v2.6.0, and iTOL v6.

Data, Materials, and Software Availability. The raw sequencing data of 10 metagenomic and 10 metatranscriptomic datasets were deposited in the National Center for Biotechnology Information (NCBI) Sequence Read Archive (SRA) under the BioProject of PRJNA1255149 (46) and PRJNA1255167 (47), respectively. All other data are included in the manuscript and/or *SI Appendix*.

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